Week 8 Homework

1 Week 8 Homework

In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred (*) proof problems are eligible for inclusion in the proof portfolio assignment.

1.1 Exercises

1.1.1 Exercise

What is Reve's Puzzle? Describe it, and describe the Frame-Stewart Algorithm in general and using an example.

1.1.2 Exercise

Use generating functions to solve:

1. $a_n = 6a_{n-1} - 8a_{n-2} + 3$ with $a_0 = 1$ and $a_1 = 0$
2. $a_n = 3a_{n-1} + 4^{n-1}$ and $a_0 = 1$

1.1.3 Exercise

Use generating functions to solve:

1. $a_n = \frac{1}{2}a_{n-1} + 1$ with $a_1 = 1$
2. $b_n = 4b_{n-1} - 3b_{n-2} + 2^n$ with $b_1 = 1$ and $b_2 = 11$

1.1.4 Exercise

Find the closed form solution to $a_n = a_{n-1} + 2(n-1)$ with $a_1 = 2$.

1.1.5 Exercise

Solve the following recurrence relations:
Find a recurrence relation for the number, t_n of length n ternary strings with no two consecutive 1s. What are the initial conditions? What is the generating function for t_n?

1.1.6 Exercise

Solve the following recurrence relations:
Find a recurrence relation for the number, d_n of length n ternary strings with no two consecutive 1s or no two consecutive 2s. What are the initial conditions? What is the generating function for d_n? (Though very similar to the previous problem, this version is much harder. Hint: The solution is not $d_n = d_{n-1} + 2d_{n-2}$, that is only part of the solution.)

1.1.7 Exercise

Consider tiling a $2 \times n$ board with 2×1 and 2×2 tiles.

1. Let d_n count the number of tilings of where the 2×1 can only be positioned vertically and 2×1 cannot be adjacent. Find 2 recursive relationship for d_n and find initial conditions for d_n.

b. Let g_n count the number of tilings of where the 2×1 can only be positioned vertically and 2×2 cannot be adjacent.

3. Find a recursive relationship for g_n and find initial conditions for g_n.

1.1.8 Exercise

Let g_n count the number of subsets of $\{1, 2, 3, \ldots, n\}$ containing no two consecutive integers. For example, $g_2 = 3$, the empty set, $\{1\}$, and $\{2\}$. Find a recurrence relationship and generating function for g_n.

1.1.9 Exercise

Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$. Find the solution when $a_0 = 1$ and $a_1 = 2$.

What do the initial terms need to be in order for $a_9 = 30$?
For which x are there initial terms which make $a_9 = x$?

1.1.10 Exercise

Consider the recurrence relation $a_n = 4a_{n-1} - 4a_{n-2}$. Find the general solution to the recurrence relation (beware the repeated root).

Find the solution when $a_0 = 1$ and $a_1 = 2$.
Find the solution when $a_0 = 1$ and $a_1 = 8$.

1.1.11 Exercise

Define the Lucas numbers as $L_n = L_{n-1} + L_{n-2}$ with $L_1 = 1$ and $L_2 = 3$. What is the generating function for this sequence? Find a closed from solution for L_n.

1.1.12 Exercise

Let $f_n = f_{n-1} + f_{n-2}$ with $f_0 = 0$ and $f_1 = 1$. Define $F_n = F_{n-1} + F_{n-2} + f_n$ for $n \geq 1$. Express F_n in terms f_n and f_{n+1}.

1.2 Proofs

1.2.1 Proof

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_0 = 0$ and $f_1 = 1$. Prove $f_0 + f_1 + f_2 + f_3 + \cdots f_n = f_{n+2} - 1$.

1.2.2 Proof

Consider the sequence $d_n = (n-1)(d_{n-1} + d_{n-2})$ with $d_1 = 0$ and $d_2 = 1$. Prove that $d_n > (n-1)!$ for $n \geq 4$.

1.2.3 Proof

Prove $\sum_{k=0}^{n} \binom{n}{k} 4^k = 5^n$.

1.2.4 Proof *

Suppose that a particular real number x has the property that $x + \frac{1}{x}$ is an integer. Prove that $x^n + \frac{1}{x^n}$is an integer for all natural numbers n.

1.2.5 Proof *

Let T_n be the triangular numbers defined by $T_n = T_{n-1} + n$ with $T_0 = 0$ and n, p, and q be positive integers. Prove that if $T_n = pq$ then $T_{n+p+q} = T_{n+p} + T_{n+q}$.

1.2.6 Proof *

Let T_n be the triangular numbers defined by $T_n = T_{n-1} + n$ with $T_0 = 0$ and n, p, and q be positive integers. Prove that if $T_{n+p+q} = T_{n+p} + T_{n+q}$ then $T_n = pq$.

1.2.7 Proof *

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_1 = 1$ and $f_2 = 2$. Calculate $f_1 + f_3 + f_5 + f_7 + \cdots + f_{2n-1}$ for small n to create a hypothesis for the sum of the first n odd indexed Fibonacci numbers, then prove this formula.

1.2.8 Proof *

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_1 = 1$ and $f_2 = 2$. Find (like the preceding problem) and prove formulas for:
$f_0 + f_2 + f_4 + f_6 + \cdots f_{2n}$

1.2.9 Proof *

Consider the sequence $f_n = f_{n-1} + f_{n-2}$ with $f_1 = 1$ and $f_2 = 2$. Find (like the preceding problem) and prove formulas for:
$f_1 - f_2 + f_3 - f_4 + \cdots (-1)^{n-1} f_n$

1.2.10 Proof *

Consider the sequence $d_n = 2d_{n-1} + d_{n-2}$ with $d_0 = 1$ and $d_1 = 3$. Prove

$$d_n = \sum_{i \geq 0} 2^i \binom{n+1}{2i}$$

is a solution to this recurrence relationship.

1.2.11 Proof *

Using the definition for $\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$ when $n < 0$, prove

$$\binom{-1/2}{n} = \frac{\binom{2n}{n}}{(-4)^n}$$

1.2.12 Proof *

Let F_n be the Fibonacci sequence with $F_1 = 1$ and $F_2 = 1$. Prove $F_n \leq (\frac{5}{3})^n$ for all positive integers.

1.2.13 Proof *

The indicated diagonals of Pascal's triangle sum to Fibonacci numbers. Prove this patterns continues forever.

Fibonacci and Pascal